S&S Quiz-10 solution

Sol (1):- Given a continious-time LTI system's difference eqn,

$$\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t) \quad -(1)$$

(a) As we know $x(t) \underset{\text{inverse}}{\overset{\mathcal{L}}{\rightleftharpoons}} X(s)$ with ROC = R where $s = \sigma + j\omega$

& $\boxed{X(s) \triangleq \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt}$ — (2)

Property:- Differentiation in the time domain

(1) mark

$$\frac{d}{dt}x(t) \overset{\mathcal{L}}{\longleftrightarrow} sX(s), \text{ with ROC containing R}$$

& $$\frac{d^n}{dt^n}x(t) \overset{\mathcal{L}}{\longleftrightarrow} s^n X(s), \text{ with ROC containing R}$$

Taking Laplace on both sides in equation (1), we get

$$s^2Y(s) - sY(s) - 2Y(s) = X(s) \quad -(3)$$

$$Y(s)[s^2 - s - 2] = X(s)$$

& $\boxed{H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 - s - 2}}$ — (5) — 1 mark

$$H(s) = \frac{1}{(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-2} \quad -(6)$$ — Partial fraction

$$\Rightarrow 1 = A(s-2) + B(s+1) \quad -(4)$$

Put $s = 2$, we get

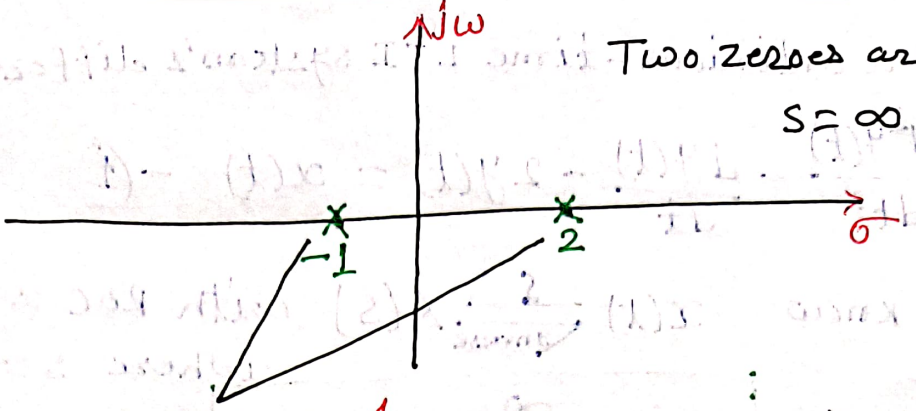
$$\Rightarrow 1 = A(2-2) + B(2+1)$$
$$= 0 + 3B = 3B$$
$$\Rightarrow B = 1/3$$

Now, Put $s = -1$

$$\Rightarrow 1 = A(-1-2) + B(-1+1)$$
$$\Rightarrow 1 = A(-3) + B \times 0 = -3A$$
$$\text{So, } A = -1/3$$

$A = -1/3$ & $B = 1/3$

Sketching Pole-zero Pattern



Two zeroes are sitting at $s = \infty$ — $\frac{1}{2}$ mark

Poles — $\frac{1}{2}$ mark

$$H(s) = \frac{\frac{1}{3}}{s-2} - \frac{1/3}{s+1} \quad -(8)$$

— by putting value of A & B in equation (6), we get

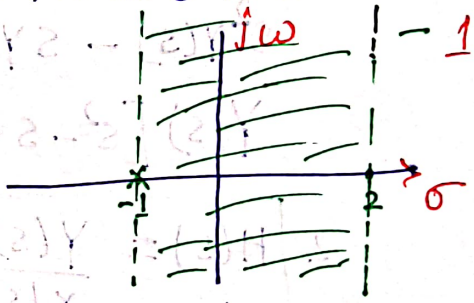
→ 1 mark

(b) <i> Stability

→ An LTI system is stable if & only if the ROC of its system function H(s) includes the entire jω axis (i.e Re(s)=0)

This will only be possible if the ROC for H(s) has to be

$$-1 < \sigma < 2$$



— 1 mark

Therefore

$$H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$$

Taking Inverse Laplace Transform

$$h(t) = -\frac{1}{3}e^{2t}u(-t) - \frac{1}{3}e^{-t}u(t)$$

$$\left[e^{-\alpha t}\cdot u(t) = \frac{1}{s+\alpha},\ \sigma > -\alpha \qquad -e^{-\alpha t}\cdot u(-t) = \frac{1}{s+\alpha},\ \sigma < -\alpha \right]$$

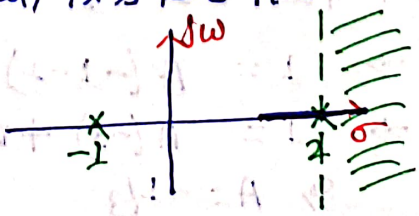
↳ 1 mark

(ii) Causality

→ For a system with a rational system function, causality of the system is equivalent to the ROC being the right-half plane to the right of the rightmost pole.

→ For system should be casual, it's ROC has to be Re(s) > 2

So, ROC $\sigma > 2$



→ 1 mark

[Right most Pole is s = 2]

So RoC $\sigma > 2$ $H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$

Therefore

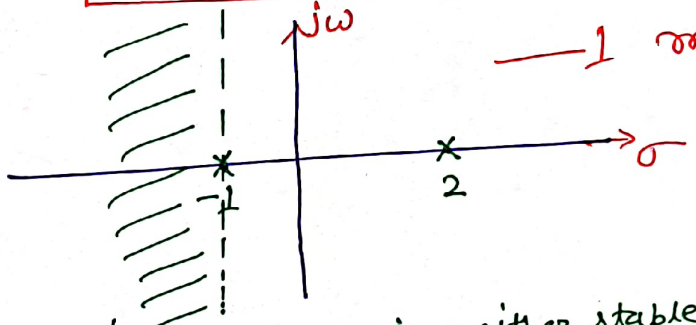
$h(t)$ after taking inverse Laplace transform is

$$h(t) = \frac{1}{3} e^{2t} u(t) - 1/3\, e^{-t} u(t)$$ — 1 mark

(iii) Neither stable nor causal

→ An LTI system is neither stable nor causal if the RoC does not contain the jw axis & it also does not have RoC, right to the right most Pole.

So RoC has to be $\sigma < -1$ & $Re(s) < -1$



— 1 mark

↳ This Region is neither stable nor causal

So, $H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$

Taking inverse Laplace Transform

$$h(t) = -1/3 \cdot e^{2t} u(-t) - \left(-1/3\, e^{-t} u(-t)\right)$$

$$h(t) = -1/3 \cdot e^{2t} u(-t) + 1/3 \cdot e^{-t} u(-t)$$ — 1 mark

Proof of $x(t) = e^{-at} u(t)$ $\quad e^{-at} u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+a}$

$$X(s) = \int_{-\infty}^{\infty} e^{-at} \cdot u(t) \cdot e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-at} \cdot e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-(s+a)\cdot t}\, dt = \frac{-\left[e^{-(s+a)\cdot t}\right]_0^{\infty}}{(s+a)}$$

$$= \frac{[e^{-\infty} - e^{0}]}{-(s+a)} = \frac{+1}{+(s+a)} = \frac{1}{s+a} \quad \sigma > -a$$

$-\frac{1}{2}$ if not done